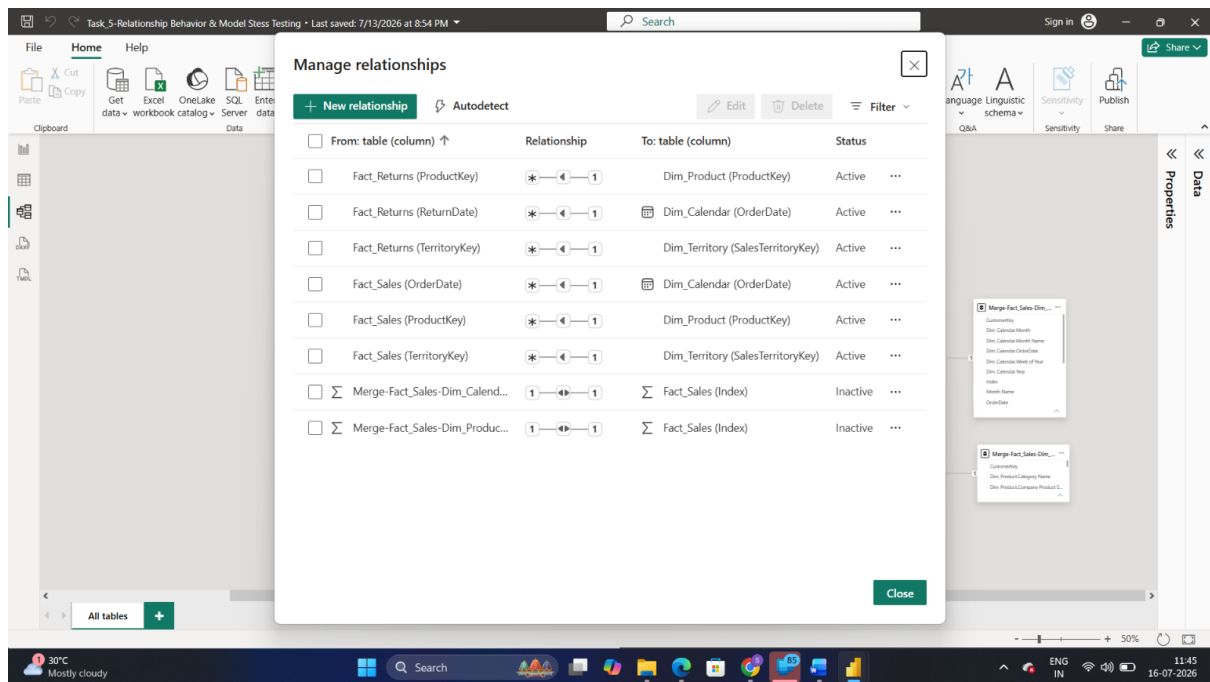


Task 5: Relationship Behaviour and Model Stress Testing

Part 1: Active & Inactive Relationships

Tasks

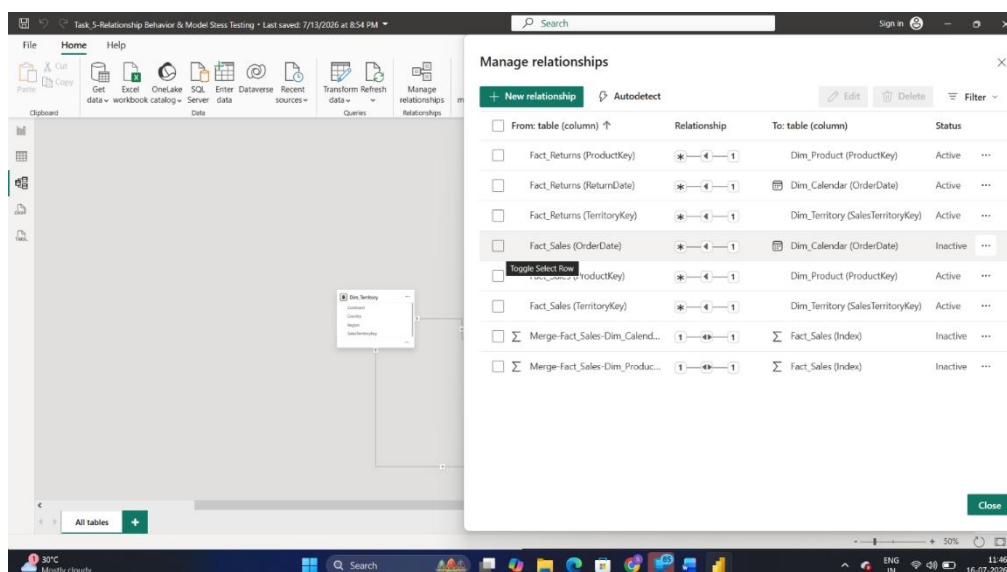
1. Identify all active relationships in the model.



2. Select one relationship between:

- Fact_Sales → Dim_Calendar

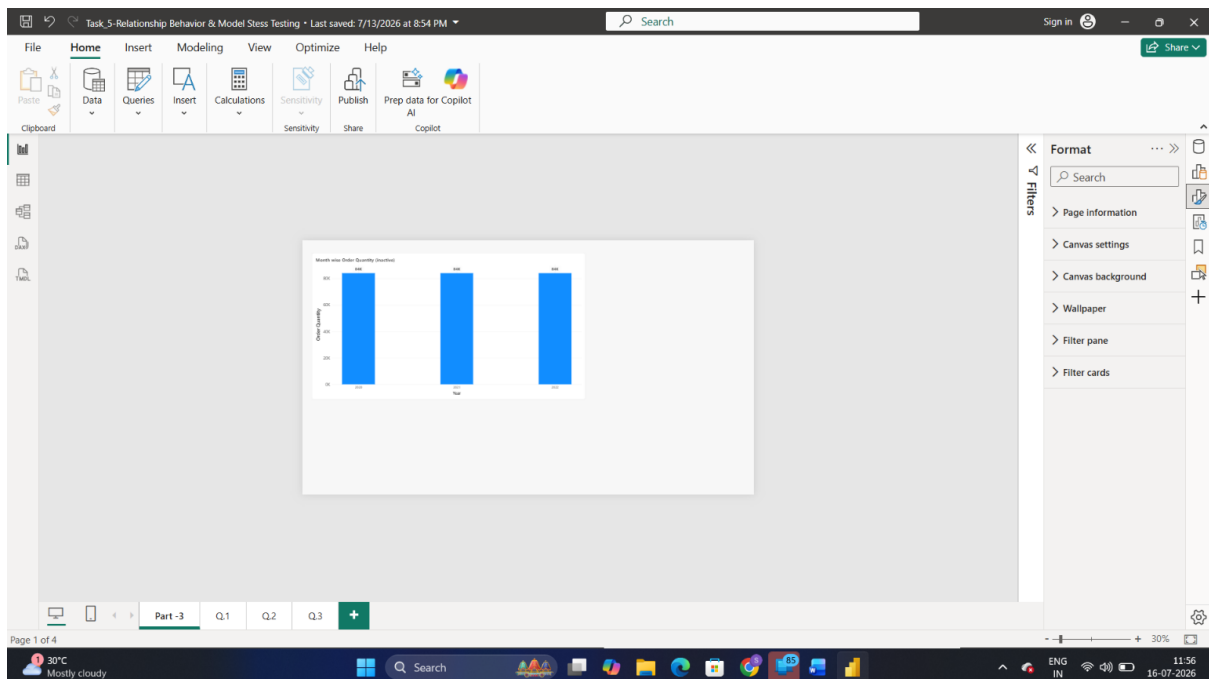
3. Make the relationship **Inactive**



Task 5: Relationship Behaviour and Model Stress Testing

4. Create a simple visual:

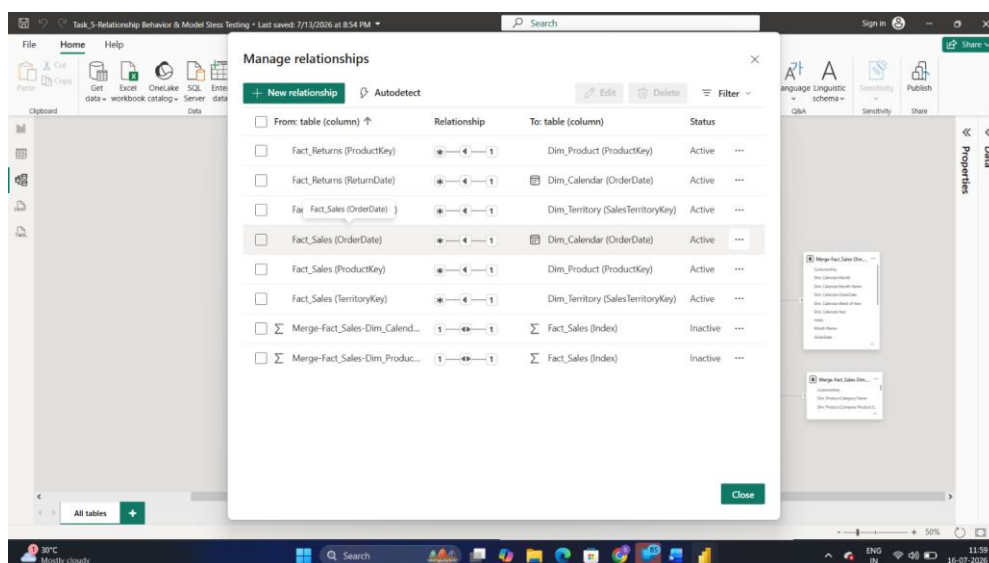
- Year vs OrderQuantity or ReturnQuantity



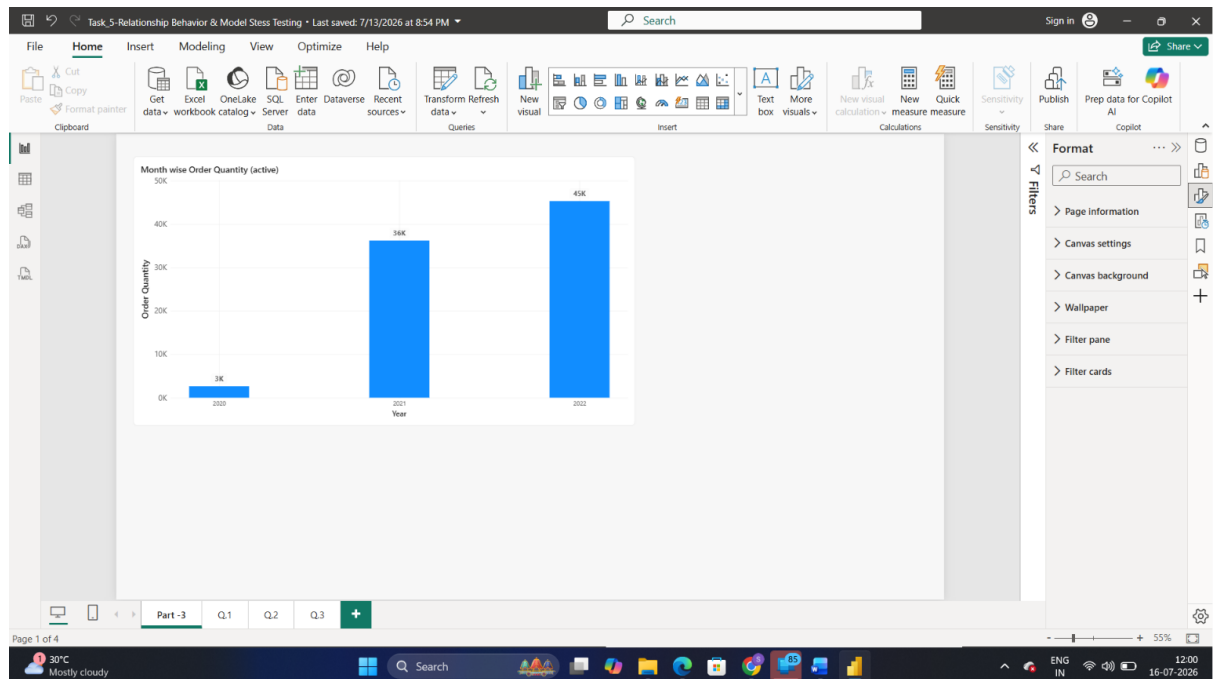
5. Observe the impact on results.

The visual was no longer being filtered through the Calendar dimension. The inactive relationship prevented normal filter propagation between the Calendar and Fact table, which can result in inaccurate report outputs.

6. Restore the relationship back to **Active**.



Task 5: Relationship Behaviour and Model Stress Testing



Analytical Questions

Q1. What change did you observe in the visual when the relationship became inactive?

After making the relationship inactive, the visual did not filter through the Dim_Calendar table as expected. The chart continued to display values, but the relationship between the Calendar dimension and the Fact table was no longer being used for filter propagation. This demonstrates that inactive relationships can prevent proper filtering and may lead to unexpected or misleading results in reports.

Q2. Why can inactive relationships cause incorrect reporting if not handled properly?

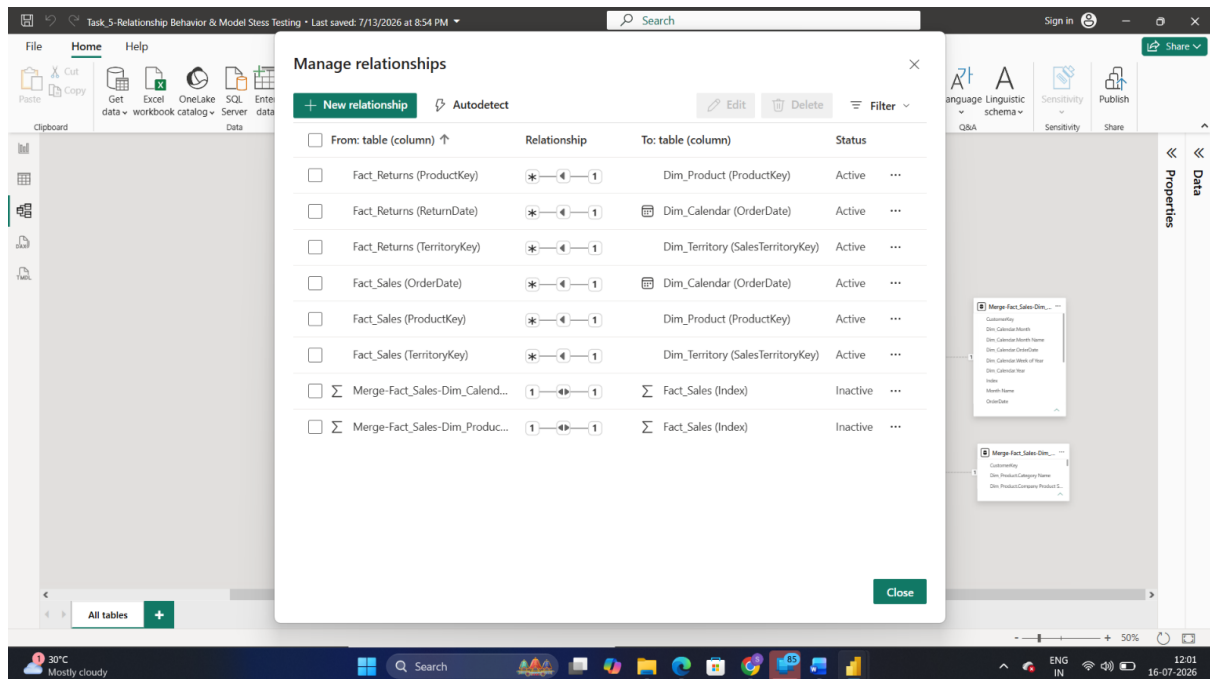
Inactive relationships block automatic filter propagation between tables. This can cause visuals to display incorrect or incomplete results, leading to inaccurate reporting and business decisions.

Part 2: Relationship Cardinality Validation

Tasks

1. Review cardinality of all relationships:
 - One → Many
 - Many → One

Task 5: Relationship Behaviour and Model Stress Testing



Analytical Question

Q3. Why must Fact → Dimension relationships always be Many → One?

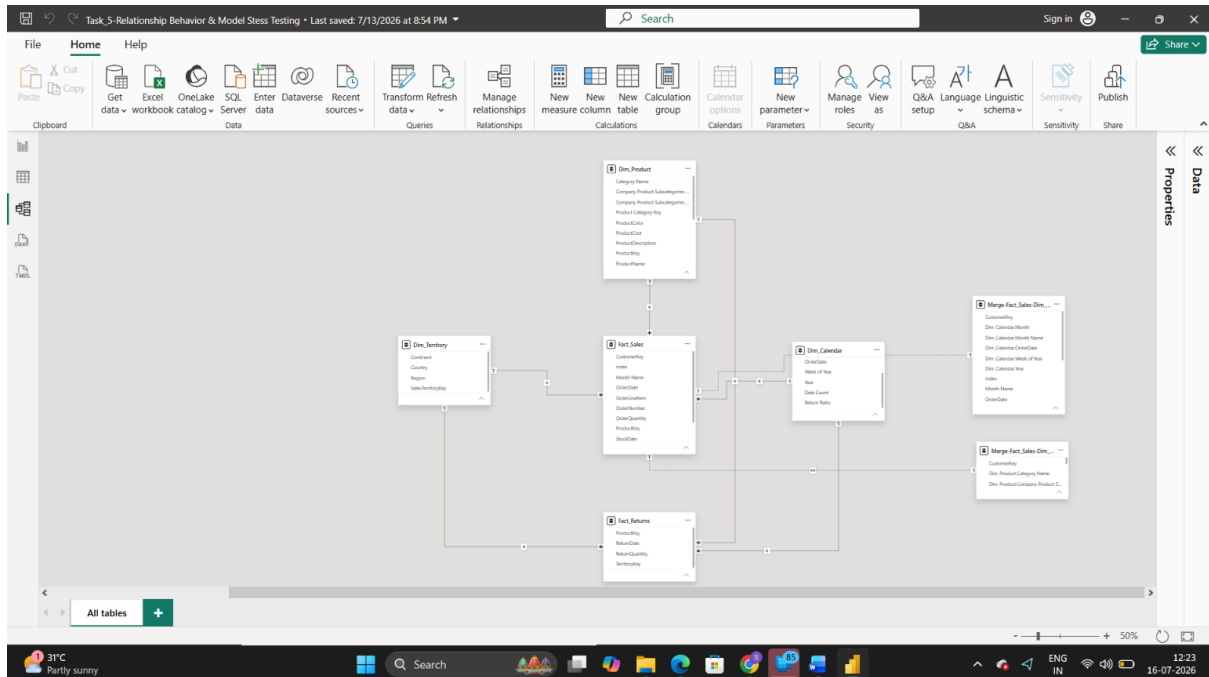
Fact tables store transactional data and therefore contain many records with repeated keys, such as ProductKey, DateKey, or TerritoryKey. Dimension tables store descriptive information and contain unique values for each key. As a result, many records in a fact table are related to a single record in a dimension table, making the correct relationship cardinality **Many-to-One**. This structure ensures accurate filtering, aggregation, and efficient reporting in a Star Schema.

Part 3: Connecting Multiple Fact Tables

Tasks

1. Observe how Fact_Sales and Fact_Returns are connected indirectly via shared dimensions.
2. Confirm that no direct relationship exists between fact tables.

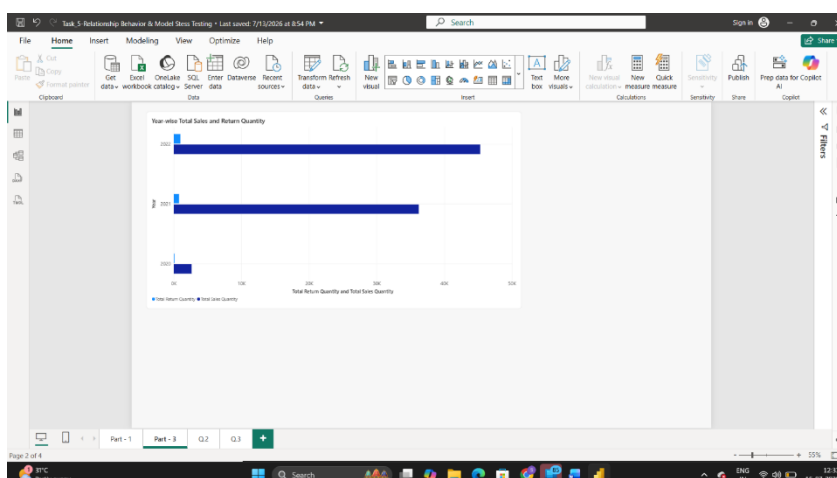
Task 5: Relationship Behaviour and Model Stress Testing



The model follows a proper Star Schema design in which **Fact_Sales** and **Fact>Returns** are not directly related to each other. Instead, both fact tables are connected through the shared dimension tables **Dim_Product**, **Dim_Calendar**, and **Dim_Territory**. This structure enables consistent analysis of sales and return data across common dimensions such as Product, Year, and Territory while preventing ambiguity, duplicate counting, and circular filter paths.

3. Create a visual displaying:

- Sales Quantity
- Return Quantity
- by Year or Category.



Analytical Question

Task 5: Relationship Behaviour and Model Stress Testing

Q4. Why should fact tables never be directly connected?

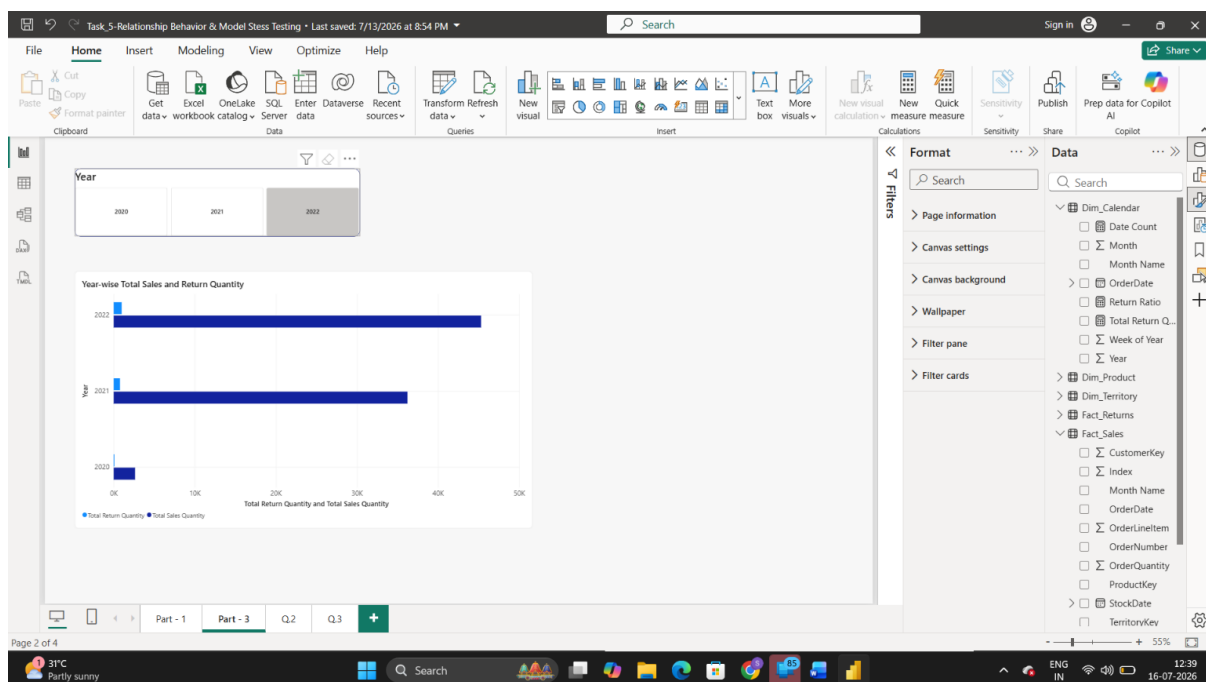
Fact tables should not be directly connected because they store transactional data and may contain different levels of detail (granularity). Direct relationships between fact tables can create ambiguity, duplicate counting, circular filter paths, and inaccurate calculations. Instead, fact tables should be connected through shared dimension tables, which provide a consistent filtering structure and ensure accurate reporting. This approach follows Star Schema best practices and improves model scalability, performance, and maintainability.

Part 4: Filter Context & Filter Flow

Tasks

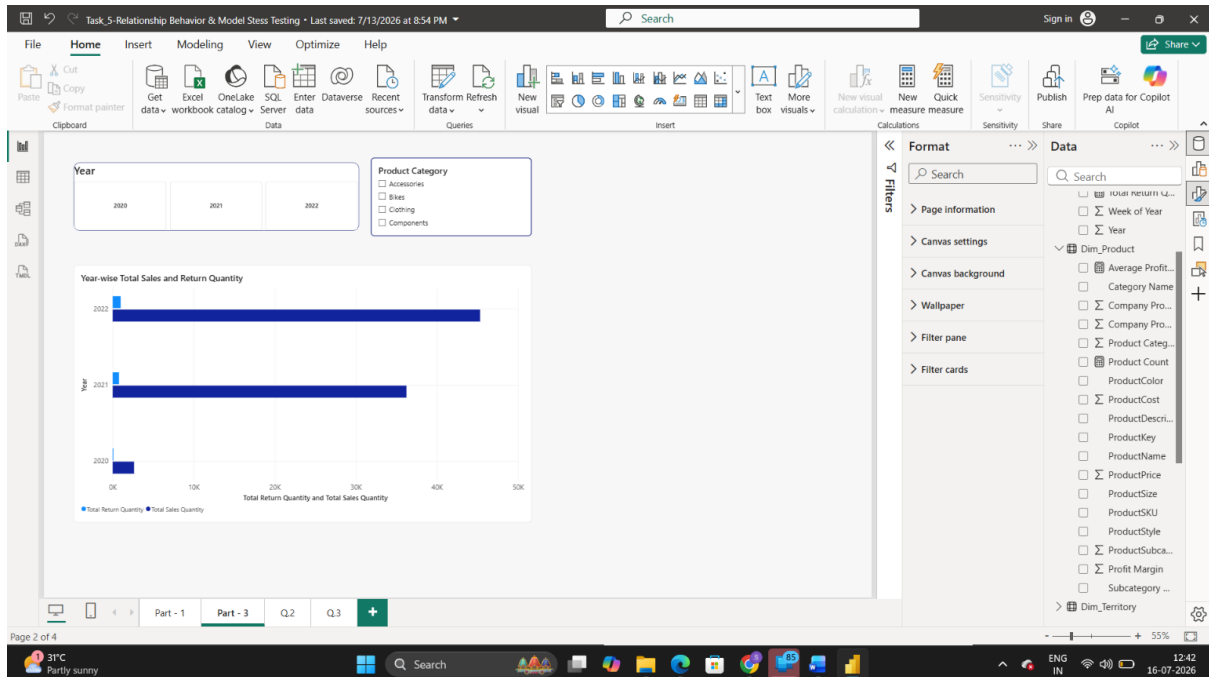
1. Use slicers for:

- Year

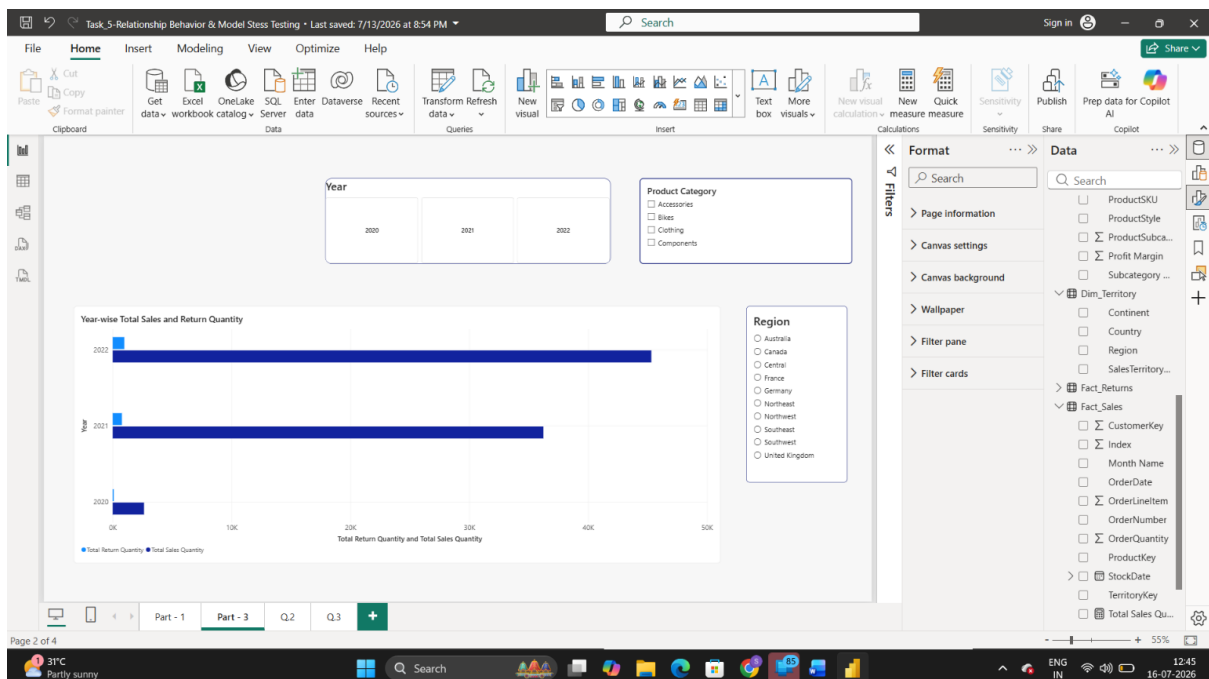


- Product Category

Task 5: Relationship Behaviour and Model Stress Testing



○ Territory

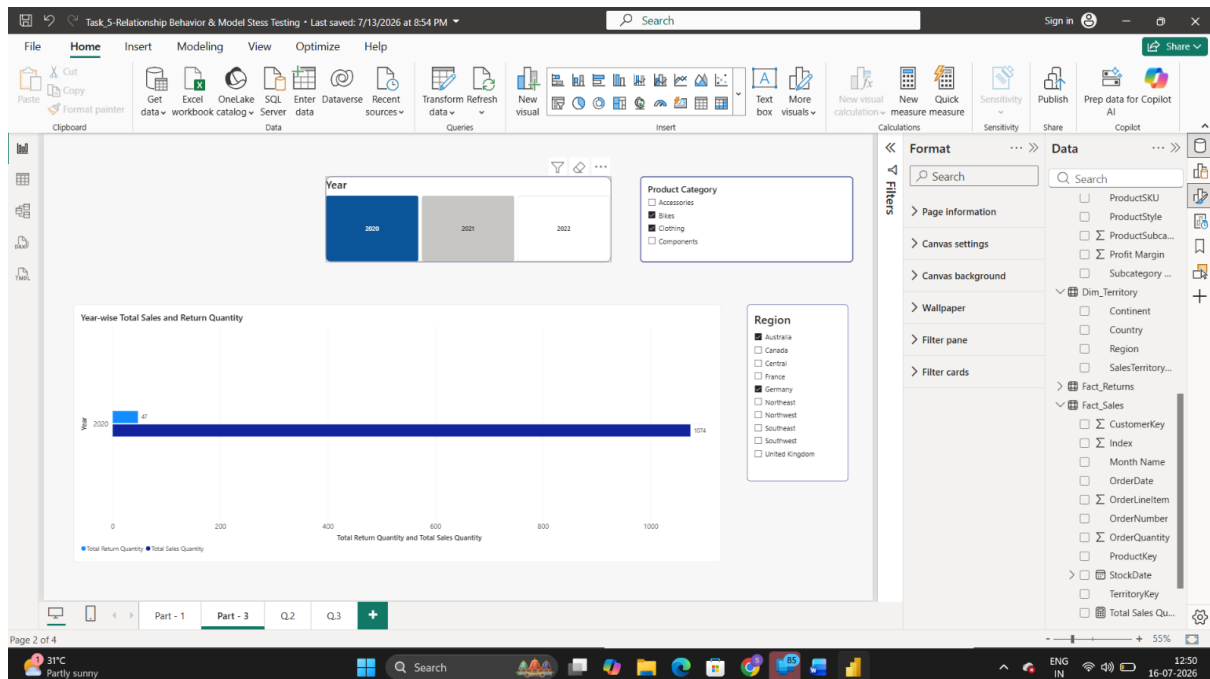


2. Observe how filters propagate from Dimension → Fact.

When slicers such as **Year**, **Product Category**, and **Territory** were applied, the visuals updated automatically to display only the corresponding Sales and Return records. This confirms that filters propagated from the Dimension tables (**Dim_Calendar**, **Dim_Product**, and **Dim_Territory**) to the Fact tables (**Fact_Sales** and **Fact_Returns**).

Task 5: Relationship Behaviour and Model Stress Testing

The filter context was correctly applied across the model, resulting in accurate and relevant aggregations.



3. Validate that filters do not incorrectly flow from Fact → Dimension.

The model was validated to ensure that filters did not flow from Fact tables back to Dimension tables. Selecting or aggregating data from **Fact_Sales** or **Fact>Returns** did not alter the available values in the Dimension tables. This confirms that the relationships follow the recommended single-direction filter flow from **Dimension** → **Fact**, preventing ambiguity and ensuring consistent reporting behavior.

Analytical Question

Q5. Explain filter flow direction in a Star Schema.

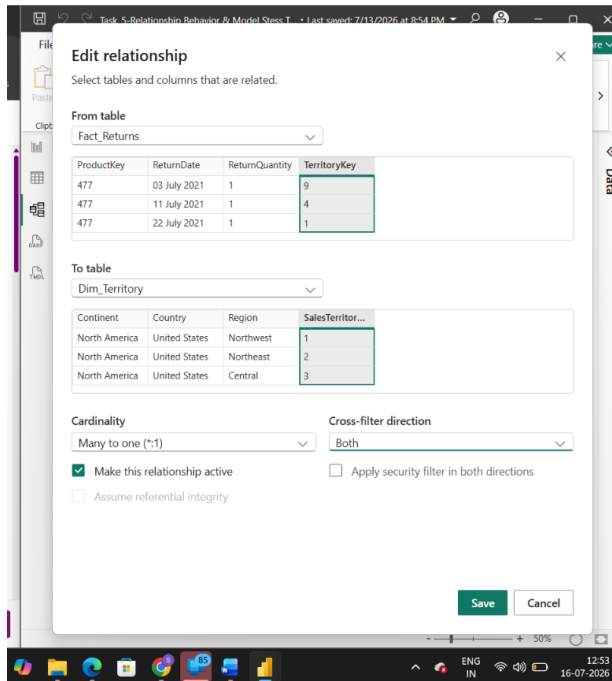
In a Star Schema, filters flow from **Dimension tables** to **Fact tables** through active relationships. When a user selects a value in a slicer, such as **Year**, **Product Category**, or **Territory**, the corresponding Dimension table filters the related records in the Fact tables. This ensures that calculations and visualizations display only the relevant transactional data. The filter direction is typically **single-directional (Dimension → Fact)**, which helps avoid ambiguity, improves performance, and ensures accurate reporting.

Part 5: Bidirectional Filter & Ambiguity

Tasks

Task 5: Relationship Behaviour and Model Stress Testing

1. Enable bidirectional filtering on one relationship.



2. Observe:

Any ambiguity warnings

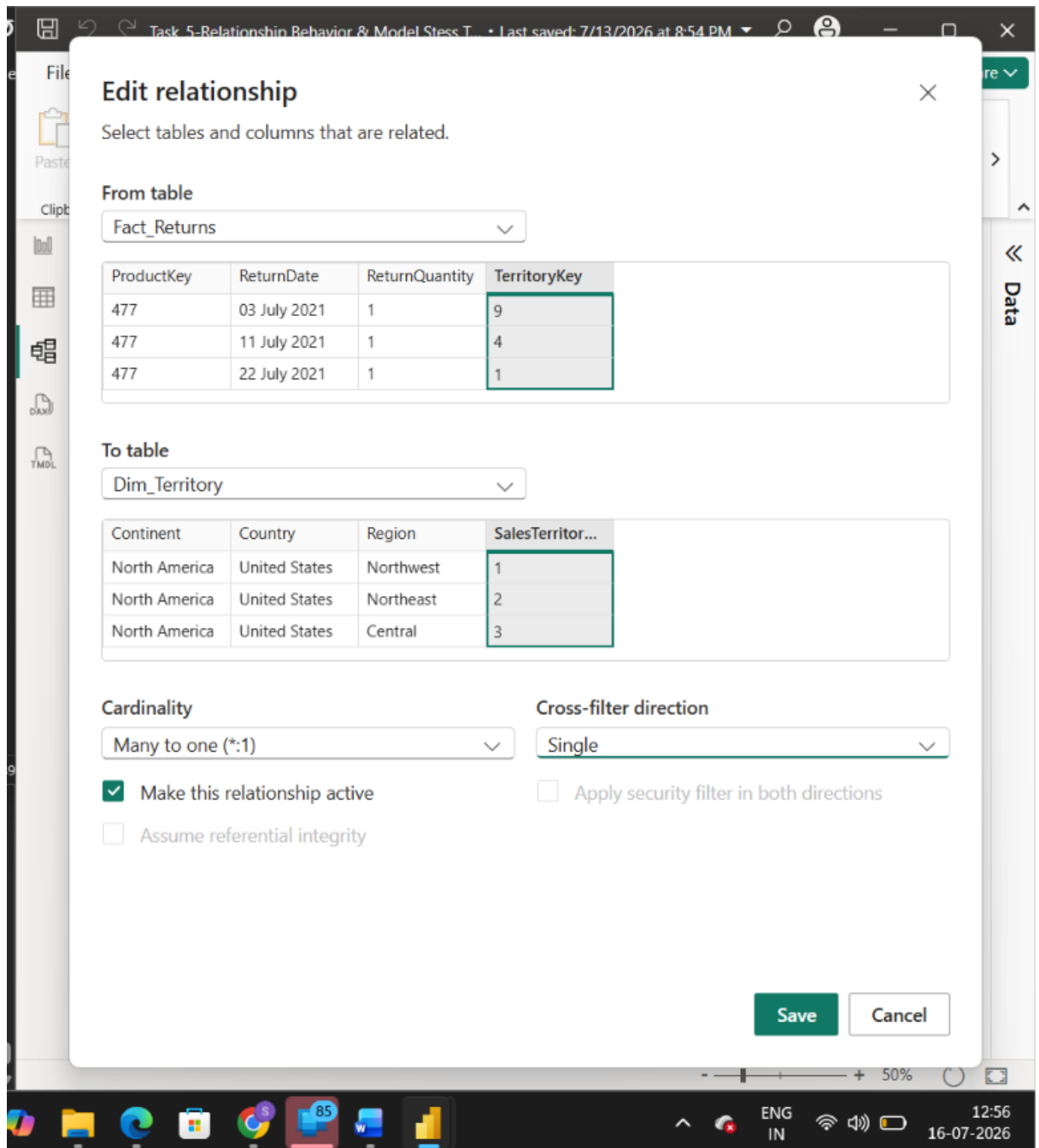
No ambiguity warning was displayed because the model contains a relatively simple Star Schema with clear relationship paths.

Unexpected visual behavior

After enabling bidirectional filtering, filters were able to propagate in both directions between the Dimension and Fact tables. Although no major issue was observed in this model, bidirectional filtering can increase model complexity and may lead to ambiguous filter paths in larger datasets.

3. Disable bidirectional filtering and restore default settings.

Task 5: Relationship Behaviour and Model Stress Testing



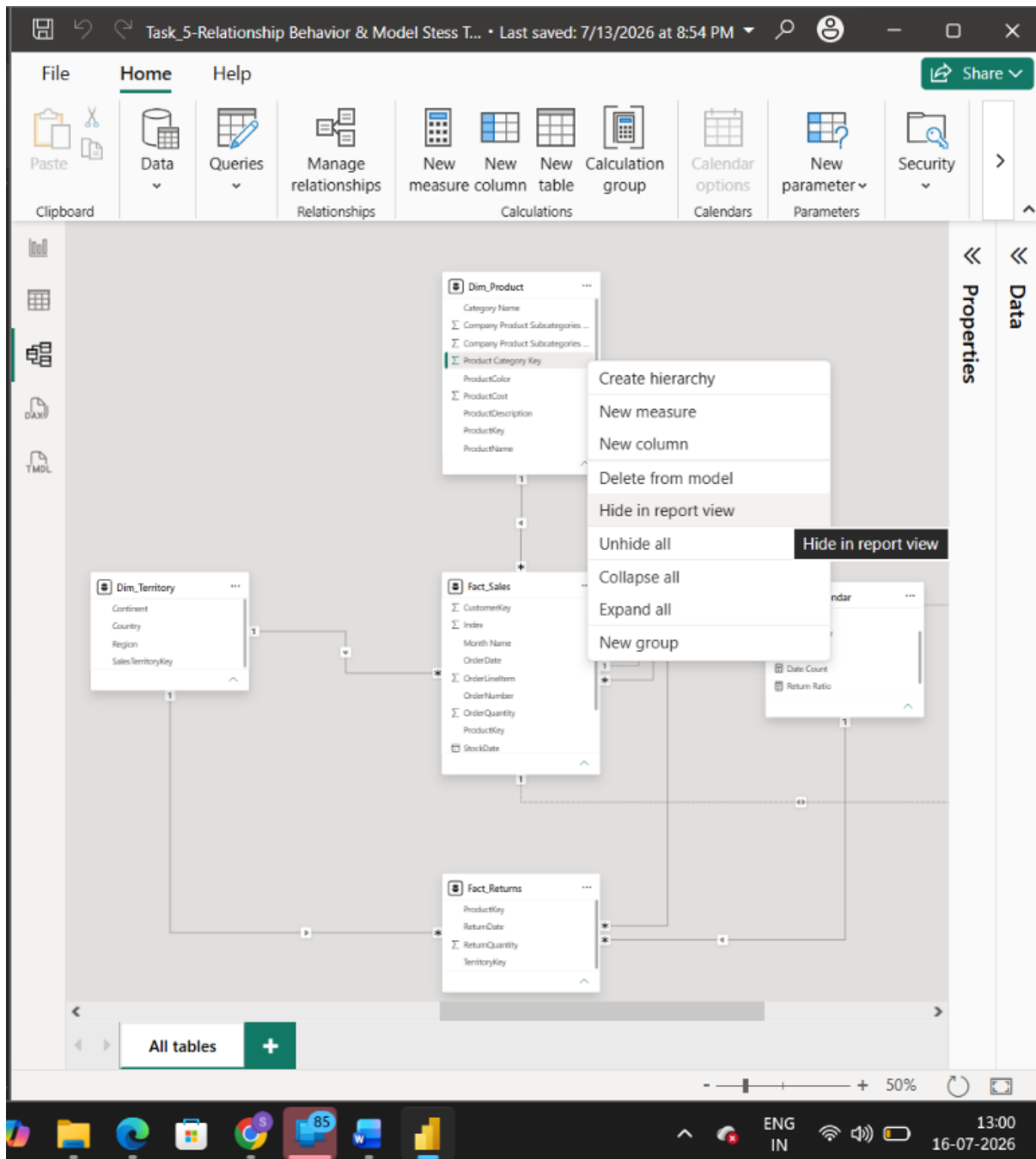
Part 6: Model Cleanup (Hide in Report View)

Tasks

1. Identify columns that should not be visible to report users, such as:
 - Keys
 - IDs

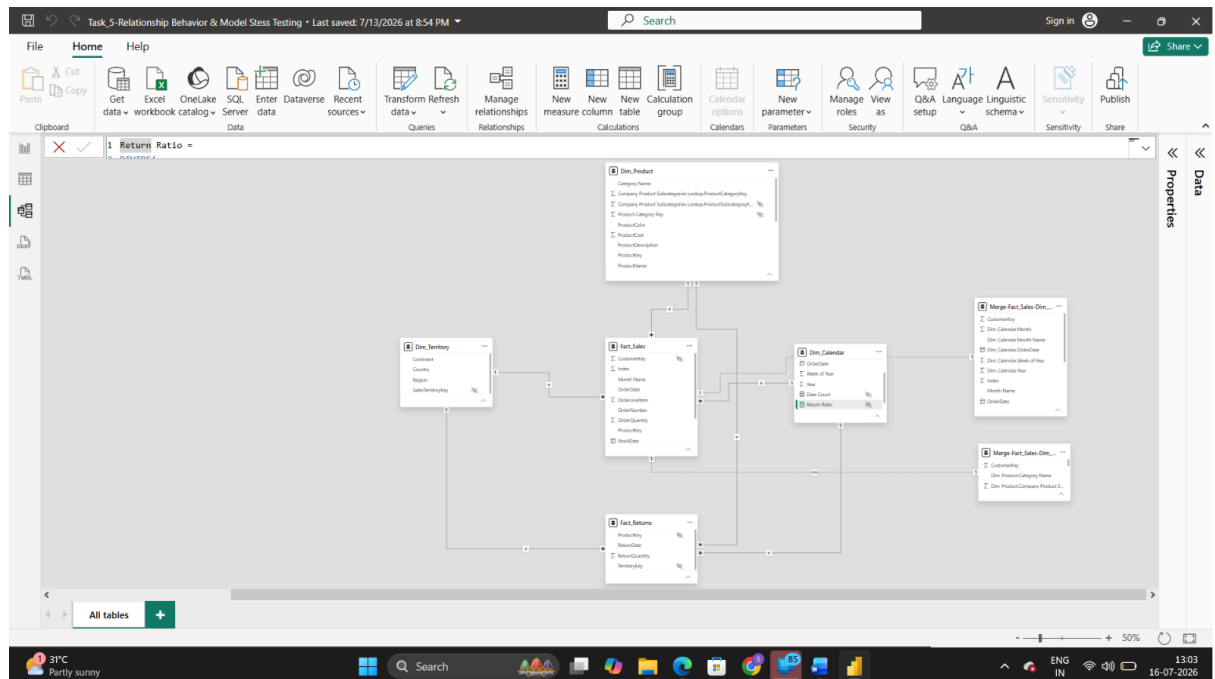
Task 5: Relationship Behaviour and Model Stress Testing

- Technical columns



2. Hide these columns in Report View.

Task 5: Relationship Behaviour and Model Stress Testing



Analytical Question

Q6. Why is hiding technical columns considered a best practice?

Hiding technical columns such as keys, IDs, and relationship fields is considered a best practice because these columns are used for data modeling and do not provide meaningful business insights to report users. Keeping them visible can clutter the Fields pane, confuse users, and increase the risk of selecting incorrect fields in reports. By hiding technical columns, the model becomes cleaner, easier to navigate, and more user-friendly, allowing report developers and business users to focus only on relevant analytical fields and measures. This improves usability, reduces errors, and enhances the overall reporting experience.
